

CSR201:STATISTICAL METHODS AND EXPLORATORY DATA ANALYSIS

L:3 T:0 P:2 Credits:4

Course Outcomes: Through this course students should be able to

CO1 :: understand basics of python and apply data manipulation techniques using NumPy and Pandas.

CO2 :: explain statistical concepts and probability distributions to support informed data-driven decision-making.

CO3 :: identify and apply effective methods for data collection and cleaning to prepare high-quality datasets for analysis.

CO4 :: apply test statistical hypotheses using appropriate inference techniques.

CO5 :: perform exploratory data analysis by utilizing statistical summaries uncover data patterns and insights.

CO6 :: apply data transformation and feature engineering to enhance predictive modeling.

Unit I

Basics of Python : Python programming basics: Variables, operators, control structures, Data Structures in Python - Lists, Tuples, Dictionaries and Sets, Functions in python, Lambda expressions, Data manipulation with Pandas, NumPy, Dataframes

Unit II

Foundations of Statistics and Data Science : Descriptive statistics: Central tendency, variance, standard deviation, Measure of dispersion, Probability fundamentals: Population vs. sample, sampling types, laws of probability, Binomial distribution, normal distribution, Introduction to data science and data lifecycle, Roles (analyst, scientist, engineer), importance of data quality, Python environment setup: IDE installation and basics.

Unit III

Data Collection, Integration, and Cleaning : Data sources and types: Structured, unstructured, semi-structured data, Data collection methods: Surveys, APIs, web scraping basics using BeautifulSoup, Data integration: Merging, joining, handling CSV, JSON, SQL formats, Data cleaning: Identifying missing data, imputation methods, handling duplicates, detecting and treating outliers using IQR and Z-score, Visualization of Outliers using Box Plots

Unit IV

Statistical Inference and Hypothesis Testing : Hypothesis formulation, null and alternative hypotheses, p-values, confidence intervals, Correlation vs. causation, covariance matrix, Distributions in hypothesis testing: t-distribution, z-distribution, chi-square test, ANOVA

Unit V

Exploratory Data Analysis and Visualization : Workflow of EDA: Goals, steps, best practices, Descriptive statistics revisited: Measures of Skewness and Kurtosis, distributions (normal, standard normal), Visualization tools: Matplotlib, Seaborn, Plotly Express, Visualization types: Histograms, boxplots, scatter plots, heatmaps, pair plots, time series and geographical plots

Unit VI

Data Transformation and Feature Engineering : Data scaling and normalization: Min-max scaling, standardization, Data encoding: Label encoding, one-hot encoding, Advanced transformations: Log, power transforms, Feature engineering: Introduction & Importance, Types of Features (Numerical, Categorical, DateTime), Creating New Features - Deriving Features from Existing Data, Feature Extraction from Dates (Day, Month, Year), Feature Selection - Importance, Techniques, Dimensionality Reduction - PCA

List of Practicals / Experiments:

List of practicals

- Generate synthetic datasets following various distributions (normal, Poisson, uniform), perform descriptive statistics and visualize with histograms and boxplots
- Web scrape data from online sources and perform cleaning and integration.
- Load real-world datasets (Iris, Tips, Airbnb), conduct full EDA including handling missing values, outliers, correlation analysis, and visual storytelling.

- Apply hypothesis testing on sample datasets, calculate confidence intervals, and interpret results.
- Implement feature engineering by deriving new variables, encoding categorical data, and applying dimensionality reduction.
- Build comprehensive Python scripts combining all steps from data loading, cleaning, transformation, visualization, and reporting insights.
- Case Study Analysis: Analyze a real-world dataset end-to-end — from collection and cleaning through EDA and feature engineering — to generate actionable insights.
- Plan a Trip with Airbnb Data: Use Airbnb listing data to find the best time and place to travel. Explore how price varies by location, number of guests, and amenities. Visualize price heatmaps and recommend “top 3 hidden gems for students on a budget.”
- Build a Music Mood Mapper: Use Spotify data to extract features like tempo, danceability, and energy. Engineer mood-based categories like “Workout Vibes” or “Chill Nights” and reduce dimensionality to find patterns. Create playlists backed by data.
- Discover the Science of Sleep: Use a fitness tracker dataset (like Fitbit or open-source sleep logs) to analyze how sleep duration, time to bed, and weekday vsweekend patterns impact mood or productivity. Create fun “sleep personality” profiles.
- Compare Campus Wi-Fi Speeds like a Data Pro: Collect speed test data from different hostel blocks or departments. Use boxplots and statistical summaries to compare performance, identify outliers, and suggest the “fastest Wi-Fi zones.”

Text Books: 1. MACHINE LEARNING IN DATA SCIENCE USING PYTHON by DR. R. NAGESWARA RAO, WILEY

References: 1. PRACTICAL STATISTICS FOR DATA SCIENTISTS by PETER BRUCE, ANDREW BRUCE, PETER GEDECK, SHROFF/O'REILLY